



OPEN Pre-trained multi-scale RWKV-GCN for multivariate time series forecasting

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Multivariate time series forecasting faces two key challenges: capturing intra-series temporal dependencies and inter-series spatial dependencies. However, heterogeneous cross-scale correlations and noise from unrelated series may obscure temporal patterns if spatial information is introduced too early. To address these issues, we propose the Pre-trained Multi-scale Receptance Weighted Key Value with Graph Convolutional Network (PMSRWKV-GCN), a two-stage framework that first learns clean temporal representations and then effectively exploits spatial structure. Before the self-supervised pre-training stage, we use the Fast Fourier Transform (FFT) to extract dominant periods. Guided by this analysis, we design a multi-scale time-mixing module derived from the Receptance Weighted Key Value (RWKV) model to align receptive fields with salient periodicities. During pre-training, we adopt a channel-independent (CI) strategy to prevent cross-channel interference and learn channel-specific temporal structure. During fine-tuning stage, a multi-scale Graph Convolutional Network (GCN) captures inter-series dependencies through scale-aware aggregation. Experiments on eight real-world datasets demonstrate that PMSRWKV-GCN achieves consistent improvements over representative baseline models. Ablation studies further confirm that CI pre-training strengthens temporal modeling, while the multi-scale GCN is critical for capturing strong spatial correlations. This stage-decoupled design effectively reconciles the trade-off between avoiding interference and leveraging spatial dependencies, yielding accurate and stable predictions.

Keywords Multivariate time series forecasting, Receptance weighted key value, Graph convolutional network, Self-supervised learning, Fast Fourier transform

Multivariate time series forecasting is widely applied in real-world domains, such as meteorology¹, transportation², energy³, and finance⁴. Accurate time series forecasting is critical for scheduling and dispatch in diverse fields. Traditional statistical methods are primarily used for univariate time series forecasting tasks, such as Exponential Smoothing (ES)⁵, Autoregressive Integrated Moving Average (ARIMA)⁶, and Autoregressive (AR)⁷. These statistical methods encounter difficulties in modeling multivariate time series, as they struggle to capture complex cross-series information.

Understanding both intra-series temporal dependencies and inter-series spatial interdependencies is the main challenge in multivariate time series forecasting. Recently, deep learning approaches have demonstrated strong capabilities in capturing complex temporal and spatial dependencies. Recurrent Neural Networks (RNNs)^{8–10} and their variants^{11–13} have been extensively used for modeling temporal dependencies in sequential data. However, RNN-based models are prone to gradient instability—both vanishing and exploding—when processing extended sequences, thereby reducing their ability to learn long-range temporal dependencies. To address these challenges, Convolutional Neural Networks (CNNs)^{14,15} and Temporal Convolutional Networks (TCNs)^{16,17} have been introduced into the field of multivariate time series forecasting. CNNs are effective at extracting features from local time windows, whereas TCNs, through their causal convolutional structure and dilated convolutional structure, are capable of modeling long-range temporal dependencies in time series data. TCNs offer advantages in stability and computational efficiency compared to traditional RNNs. Nevertheless, both CNNs and TCNs face limitations in modeling complex inter-series relationships.

More recently, the Transformer¹⁸ architecture has gained attention for its success in Natural Language Processing (NLP), owing to its self-attention mechanism and parallel processing capabilities. Transformers

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can capture long-range dependencies within sequences and leverage multi-head self-attention to learn features from various perspectives. Many studies have applied the Transformer to time series forecasting tasks, such as Informer¹⁹, Autoformer²⁰, Reformer²¹, Crossformer²², and iTransformer²³. However, the Transformer architecture suffers from high memory usage and computational overhead when processing long sequences. To address the quadratic complexity of Transformer attention, Peng et al.²⁴ introduced RWKV, which combines the efficient recurrent inference of RNNs with the parallel training advantages of Transformers. The RWKV architecture has achieved promising performance in some research fields^{25–28}. Although RWKV is effective for modeling long sequences through recurrent weighted key–value accumulation, its time-mixing mechanism does not explicitly provide receptive field alignment across different temporal scales or dominant periods. This motivates the introduction of multi-scale temporal modeling to better capture scale-aware temporal patterns. Additionally, Graph Neural Networks (GNNs)^{29–31} have emerged as a promising tool for multivariate time series forecasting, particularly for capturing spatial dependencies among series corresponding to different sensors or geographical locations. GNNs model the relationships between different time series as a graph structure, enabling the model to predict outcomes within a complex network of relationships.

Although deep learning models have greatly advanced multivariate time series forecasting, effectively unifying temporal and spatial dependencies across multiple scales remains a nontrivial problem. Moreover, the presence of weakly correlated or irrelevant series can introduce noise, which often obscures meaningful temporal patterns and reduces predictive accuracy. To address these issues, we propose a novel PMSRWKV-GCN model, which utilizes a CI strategy during the pre-training stage. The proposed framework improves forecasting accuracy and stability for multivariate time series data through a multi-scale learning scheme that models both temporal dynamics within individual series and spatial correlations across different series. Comprehensive experiments on eight real-world datasets verify the effectiveness of PMSRWKV-GCN compared with several representative deep learning baselines. The key contributions of this work are outlined below:

1. We introduce PMSRWKV-GCN, a two-stage framework that jointly captures temporal and spatial dependencies for multivariate time series forecasting. The framework leverages the FFT to identify dominant periods and align multi-scale correlations across both the pre-training and fine-tuning stages.
2. In the pre-training stage, a CI strategy enables each time series channel to learn distinctive temporal representations without interference from others. This design effectively reduces noise from weakly correlated series and enhances the quality of temporal features.
3. During fine-tuning, a multi-scale GCN learns inter-series dependencies at different scales, enabling the framework to adaptively exploit spatial information and improve forecasting accuracy and generalization.

The remainder of this paper is structured as: Sect. 2 presents concepts of RWKV architecture and applications of GNNs in time series forecasting; Sect. 3 details the architecture of PMSRWKV-GCN; Sect. 4 shows the experimental evaluation; and Sect. 5 summarizes the paper with discussions and future directions.

Related works

The RWKV architecture

The RWKV model is an advanced neural network architecture designed to effectively model long-range temporal relationships²⁵. RWKV differs from traditional RNNs and Transformers by leveraging a novel mechanism that avoids the quadratic cost of self-attention, thus making it computationally efficient for long sequences. RWKV is composed of two key modules: a time-mixing block and a channel-mixing block.

The time-mixing block captures the temporal dependencies across different time steps using the token shift strategy. The Key, Value, and Receptance vectors are calculated using the following equations:

$$r_t = W_r \cdot (\mu_r x_t + (1 - \mu_r)x_{t-1}) \quad (1)$$

$$k_t = W_k \cdot (\mu_k x_t + (1 - \mu_k)x_{t-1}) \quad (2)$$

$$v_t = W_v \cdot (\mu_v x_t + (1 - \mu_v)x_{t-1}) \quad (3)$$

The weighted key-value computation, wkv_t , is given by:

$$wkv_t = \frac{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} v_i + e^{u+k_t} v_t}{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} + e^{u+k_t}} \quad (4)$$

$$o_t = W_o \cdot (\sigma(r_t) \odot wkvt) \quad (5)$$

where o_t is the output at time point t ; W_r , W_k , W_v , and W_o are learnable weight matrices; the parameters u_r , u_k , and u_v are scaling factors of the token shift strategy; σ is the *Sigmoid* function; w and u are scalars; and $\odot$ denotes element-wise multiplication. The function of wkv_t is similar to the $Attention(Q, K, V)$ in Transformers but maintains linear-time computation and fixed memory usage.

The channel-mixing block is designed to capture dependencies across different features. The computation involves the following:

$$r_t = W_r \cdot (\mu_r x_t + (1 - \mu_r)x_{t-1}) \quad (6)$$

$$k_t = W_k \cdot (\mu_k x_t + (1 - \mu_k)x_{t-1}) \quad (7)$$

$$o_t = \sigma(r_t) \odot (W_v \cdot \text{ReLU}(k_t)^2) \quad (8)$$

where a squared *ReLU* is used as the activation function. The *Sigmoid* activation applied to the receptance serves as a “forget gate”, helping to filter out unnecessary historical information.

The RWKV offers a novel approach to long-sequence processing by avoiding the computational pitfalls of traditional attention mechanisms, making it highly efficient. This architecture excels in capturing complex temporal dynamics across long sequences, establishing RWKV as a promising direction for advanced sequential modeling.

Applications of GNNs in time series forecasting

GNNs^{32,33} have become increasingly influential for time series analysis, particularly for scenarios where data exhibit inherent graph structures. They effectively model spatial dependencies among entities, which is essential when interactions between variables or nodes play a key role. For instance, in traffic forecasting^{34–36}, GNNs can model the interactions between different locations based on road network graphs. Popular GNN architectures such as GCNs^{37,38}, Graph Attention Networks (GATs)^{39,40}, and GraphSAGE⁴¹ have been employed to leverage the spatial structure of data. GCNs apply convolution operations over graph-structured data, allowing the model to aggregate information from neighboring nodes. GATs extend conventional GCNs by assigning adaptive weights to neighboring nodes through an attention mechanism. When applied to time series, GNNs enable the joint modeling of temporal dynamics and spatial correlations across multiple channels. GNNs have demonstrated a remarkable ability in capturing spatial correlations among multiple time series, effectively modeling interactions between related variables through graph representations. By learning dependencies embedded in these graph structures, GNN-based models can handle complex relational dynamics and often surpass conventional statistical approaches, particularly in scenarios where inter-variable interactions play a decisive role.

Beyond time series forecasting, the combination of frequency-aware modeling, spatial dependency learning, and multi-scale graph representations has also been widely explored in other research domains. For example, FSCMF⁴² employs frequency–spatial joint perception for visible–infrared image fusion, while TSMGA⁴³ utilizes temporal–spatial multi-scale graph attention for remote sensing change detection. Related ideas of decoupling motion and appearance, integrating multi-sensor spatial information, and structure-aware modeling for anomaly and outlier detection have also been investigated in event camera representation learning⁴⁴, radar–camera fusion tasks⁴⁵, and complex data analysis⁴⁶. These studies collectively demonstrate that jointly modeling frequency characteristics, spatial structures, and structural relationships is a general and effective strategy across different modalities.

Inspired by these advances, we integrate the RWKV architecture—known for its efficiency in modeling long-range temporal dependencies—with the spatial reasoning capability of GNNs. This hybrid framework leverages the complementary strengths of both paradigms, enabling accurate and stable multivariate forecasting. It not only alleviates the limitations of traditional architectures but also provides an effective approach to jointly capturing temporal and spatial dependencies with enhanced accuracy and stability.

Methodology

FFT for period identification

In domains such as photovoltaic power generation and meteorology, multivariate time series often exhibit periodic behaviors driven by natural cycles on daily, monthly, or annual scales. These recurring patterns influence system dynamics and complicate forecasting. To capture such regularities, we apply the FFT to project time-domain signals into the frequency domain, where dominant periods can be more readily identified. This frequency-based perspective enhances the model’s capacity to capture periodic structures that are otherwise difficult to identify in the time domain.

Specifically, we apply FFT to the multivariate time series data $\mathbf{X} \in \mathbb{R}^{L \times N}$, where L is the sequence length and N is the number of dimensions. The procedure is mathematically described as:

$$\mathbf{A} = \text{Avg}(\text{Amp}(\text{FFT}(\mathbf{X}))) \quad (9)$$

where $\text{FFT}(\cdot)$ represents the process of FFT, $\text{Amp}(\cdot)$ denotes the amplitude function, and $\text{Avg}(\cdot)$ represents the average operation over the N dimensions. To retain the dominant periodic information while suppressing irrelevant high-frequency noise, we select the frequency indices corresponding to the k largest amplitudes in $\mathbf{A}$. Owing to the conjugate symmetry of the frequency spectrum, only the first half of the frequency range $\{1, \dots, [L/2]\}$ is preserved:

$$\{f_1, f_2, \dots, f_k\} = \text{TopKIndex}(\mathbf{A}), f_i \in \{1, \dots, [L/2]\} \quad (10)$$

Here, k represents the number of dominant frequencies retained. The choice of k helps to mitigate the sparsity of the frequency domain representation and minimize the impact of noise from high-frequency components that may not contribute meaningful information to the forecasting model⁴⁷. These selected frequencies correspond to the most influential periodic patterns in the data, which are essential for accurate modeling and forecasting.

Each selected frequency f_i is then converted to a period using the formula:

$$p_i = \frac{L}{f_i}, i \in \{1, \dots, k\} \quad (11)$$

This identification of periods enables the subsequent modeling stages to focus on these key periodicities, thereby improving the accuracy and effectiveness of the forecasting model. By emphasizing these dominant cycles, the approach mitigates the cross-scale complexity inherent in multivariate time series, particularly in applications sensitive to periodic fluctuations.

Multi-scale time-mixing module

The standard RWKV time-mixing block adopts a linear attention mechanism that enables parallel training comparable to the Transformer architecture. However, its reliance on only one preceding state limits its ability to capture long-term dependencies. To address this, we redesign the time-mixing block into a multi-scale variant that incorporates dominant periodic cues identified by FFT, extends the effective receptive field, and captures temporal patterns across multiple resolutions. The enhanced multi-scale time-mixing module is defined as follows:

Frequency extraction

We utilize the FFT to extract significant periodicities from the time series. The top- k frequencies, $\{f_1, f_2, \dots, f_k\}$, are selected based on their corresponding amplitudes, $\{a_1, a_2, \dots, a_k\}$, which are then normalized using a *Softmax* function as follows:

$$\{a'_1, a'_2, \dots, a'_k\} = \text{Softmax}(a_1, a_2, \dots, a_k) \quad (12)$$

The normalized amplitudes serve as scaling factors for the k period levels, allowing the model to dynamically fuse information from multiple periodic components when generating subsequent representations.

It is important to note that, unlike Autoformer²⁰ and TimesNet⁴⁷ where FFT is used for explicit temporal reshaping or decomposition, in our framework FFT only serves as a guiding signal for receptive field alignment and scale-aware graph aggregation.

Multi-scale feature aggregation

For each time step t , the Key, Value, and Receptance vectors are computed by aggregating information from multiple periods. The calculation is expressed as:

$$r'_t = W_r \cdot (\mu_r x_t + (1 - \mu_r)(x_{t-1} + \sum_{i=1}^k a'_i x_{t-p_i})) \quad (13)$$

$$k'_t = W_k \cdot (\mu_k x_t + (1 - \mu_k)(x_{t-1} + \sum_{i=1}^k a'_i x_{t-p_i})) \quad (14)$$

$$v'_t = W_v \cdot (\mu_v x_t + (1 - \mu_v)(x_{t-1} + \sum_{i=1}^k a'_i x_{t-p_i})) \quad (15)$$

Here, a'_i and p_i corresponds to the weight and period derived from the selected frequencies, respectively.

Multi-head computation

To enhance the model's ability to capture diverse temporal patterns, we incorporate a multi-head computation mechanism. This mechanism involves multiple heads, each corresponding to different periods identified through FFT process. The outputs of these heads are computed as follows:

$$wkv_t^0 = \frac{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} v_i + e^{u+k_t} v_t}{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} + e^{u+k_t}} \quad (16)$$

$$wkv_t^j = \frac{\sum_{i=1}^{t-1} e^{-(t-1-p_j i)w+k_i} v_i + e^{u+k_t} v_t}{\sum_{i=1}^{t-1} e^{-(t-1-p_j i)w+k_i} + e^{u+k_t}}, j \in [1, \dots, k] \quad (17)$$

Here, wkv_t^0 corresponds to the original series, and $wkv_t^1, \dots, wkv_t^k$ correspond to the series processed with different periodic scales $p_1, \dots, p_k$; w and u are scalars. The resulting representations are concatenated to form the aggregated multi-head output:

$$\text{MultiHead}(wkv'_t) = \text{Concat}(wkv_t^0, wkv_t^1, \dots, wkv_t^k) \quad (18)$$

The output of the proposed multi-scale time-mixing module is expressed as:

$$o'_t = W_o \cdot (\sigma(r'_t) \odot \text{MultiHead}(wkv'_t)) \quad (19)$$

where σ denotes the *Sigmoid* activation. Building on the design principles of prior models such as Autoformer²⁰ and TimesNet⁴⁷, our module extends temporal mixing beyond short-term dependencies by embedding multi-scale periodic cues, thereby strengthening the representation of long-range temporal dynamics.

Multi-scale GCN module

Since inter-series dependencies in multivariate time series can vary across different temporal scales, we design a multi-scale GCN to effectively capture spatial relationships at multiple level. Figure 1 shows the architecture of the proposed multi-scale GCN module. Leveraging the graph structure, the multi-scale GCN module can capture cross-channel dependencies over multiple temporal scales, thereby enhancing forecasting accuracy in complex systems.

The first step involves dividing the input time series $\mathbf{X} \in R^{L \times N}$ into segments based on identified scales $\{p_1, p_2, \dots, p_k\} (p_i < L)$ from FFT analysis. Specifically, the time window L is segmented into $\lceil L/p_i \rceil$ parts, where each segment aligns with the periodic patterns detected. To ensure complete alignment with periodic structures, the last segment is cycle-padded using the initial portion of the sequence when necessary. This segmentation ensures that the data is represented at multiple temporal resolutions, facilitating the capture of scale-specific dependencies. For each scale p_i , the input is reshaped to fit the segmentation:

$$X^i = \text{Reshape}(\text{Padding}(\mathbf{X})), i \in \{1, \dots, k\} \quad (20)$$

For each temporal segment at a specific scale p_i , a GCN is employed to model spatial dependencies among variable. The initial adjacency matrix A is initialized with uniform weights, assuming equal inter-series correlation prior to training. The multi-scale GCN operation includes the following:

Node feature update

For each temporal segment $j \in \{1, \dots, \lceil L/p_i \rceil\}$ corresponding to a specific period scale p_i , the node features are updated using a GCN. All segments under the same scale share a common adjacency matrix A_i , ensuring consistent spatial relationships across segments, while each segment's feature matrix H_j is independently updated according to:

$$H_j^{(l+1)} = \sigma(A_i H_j^{(l)} W_j^{(l)}), j \in \{1, \dots, \lceil L/p_i \rceil\} \quad (21)$$

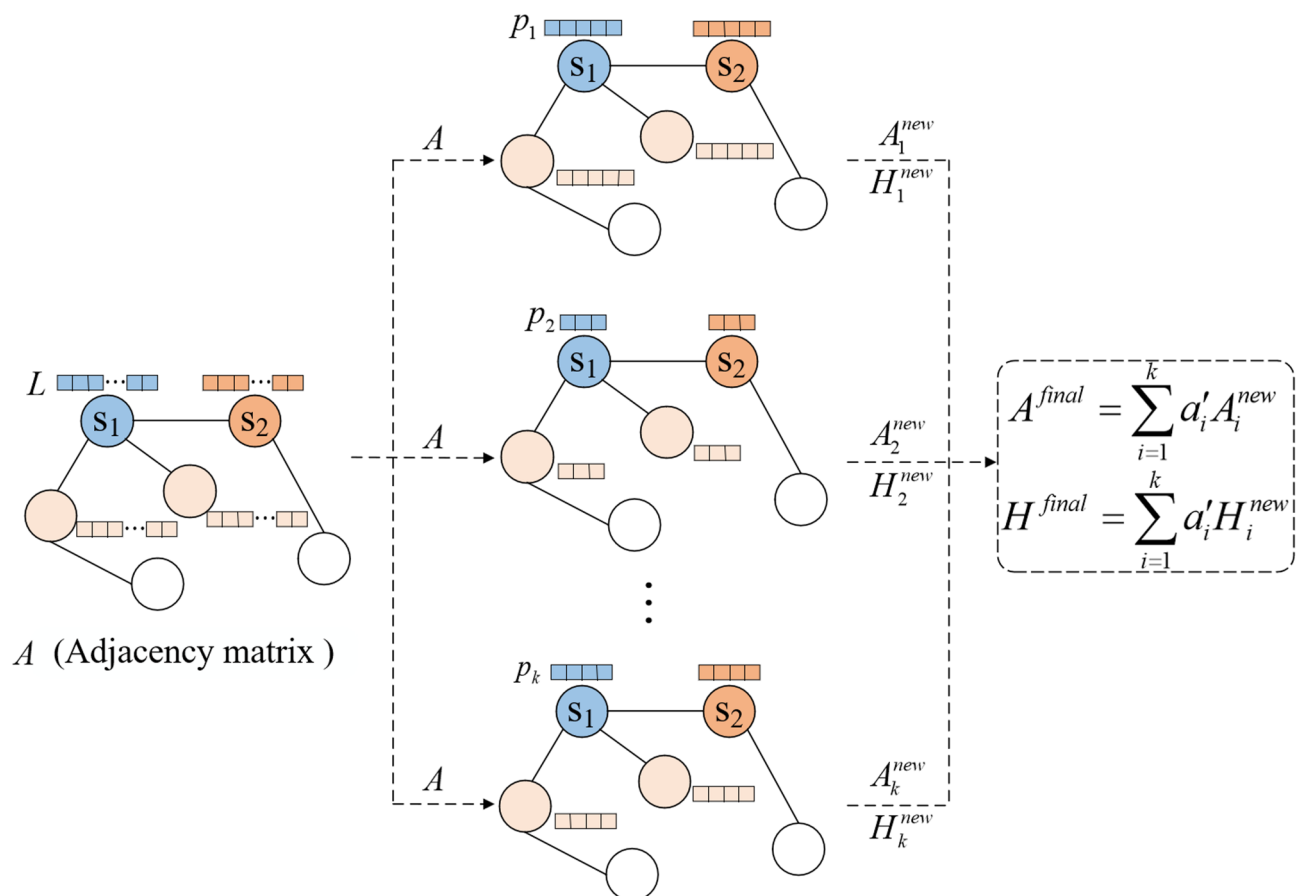


Fig. 1. Overall process of the proposed multi-scale GCN module. The input time series $\mathbf{X} \in R^{L \times N}$ is decomposed into multiple periodic scales $\{p_1, p_2, \dots, p_k\}$ identified via FFT analysis. Each scale is processed by an independent GCN to update node features H_i^{new} and adjacency matrices A_i^{new} . The outputs from all scales are subsequently aggregated using FFT-derived weights $\{a'_1, a'_2, \dots, a'_k\}$.

where $H_j^{(l+1)}$ represents the updated feature matrix at layer $l+1$, $W_j^{(l)}$ is the trainable weight matrix, and σ is the activation function.

After processing all segments corresponding to a specific period scale p_i , the outputs of these segments are concatenated along the temporal dimension to form a unified feature matrix. This concatenation aggregates information across segments, yielding a comprehensive representation of the temporal dynamics at scale p_i :

$$H_i^{new} = \text{Concat}(H_1^{(l+1)}, H_2^{(l+1)}, \dots, H_{\lceil L/p_i \rceil}^{(l+1)}), i \in \{1, \dots, k\} \quad (22)$$

Dynamic adjacency matrix update

To adaptively capture the evolving inter-channel relationships, the adjacency matrix A_i , corresponding to each period scale p_i is dynamically refined through a self-attention mechanism. This module enables the model to learn data-driven spatial dependencies rather than relying on a fixed adjacency structure. For each scale p_i , the corresponding *query*, *key*, and *value* matrices are obtained as:

$$Q_i = H_i W_Q, K_i = H_i W_K, V_i = H_i W_V \quad (23)$$

where W_Q , W_K and W_V are learnable projection matrices.

The attention scores are calculated as:

$$\text{attention} = \text{Softmax}(\text{LeakyReLU}(Q_i K_i^T / \sqrt{d})) \quad (24)$$

where the *LeakyReLU* activation introduces non-linearity into the pairwise attention computation, and the *Softmax* function normalizes the attention scores to ensure numerical stability. The resulting attention weights capture the pairwise dependencies between channels and are used to update the adjacency matrix as:

$$A_i^{new} = \text{attention} \cdot V_i, i \in \{1, \dots, k\} \quad (25)$$

This dynamic refinement enables the adjacency matrix to evolve adaptively with the data, thereby enhancing the model's ability to capture dynamic spatial correlations across scales.

Node feature aggregation

After the GCN step, each scale p_i produces a scale-specific feature matrix H_i^{new} . To integrate information across scales, we perform a weighted aggregation using the FFT-derived importance weights $\{a'_1, a'_2, \dots, a'_k\}$. The final aggregated node representation is

$$H^{final} = \sum_{i=1}^k a'_i H_i^{new} \quad (26)$$

This operation emphasizes scales with stronger periodic energy while retaining contributions from the remaining scales.

Adjacency matrix aggregation

In parallel, the scale-wise updated adjacency matrices $\{A_i^{new}\}_{i=1}^k$ are fused with the same weights to obtain a single, scale-aware graph:

$$A^{final} = \sum_{i=1}^k a'_i A_i^{new} \quad (27)$$

Using identical weights for features and graphs keeps the semantic alignment between the node representations and the relational structure learned at each scale.

By leveraging weighted fusion, the multi-scale GCN module effectively captures the most informative intra-series temporal dependencies and inter-series spatial dependencies across multiple scales, yielding a stable and globally consistent representation that enhances downstream forecasting. Furthermore, the adaptive adjacency mechanism enables the module to dynamically adapt to evolving inter-series relationships, ensuring stability under complex and non-stationary conditions.

CI pre-training for noise-free temporal representation

In recent years, self-supervised pre-training has emerged as a promising paradigm beyond traditional domains such as NLP⁴⁸ and Computer Vision (CV)⁴⁹, demonstrating strong adaptability to time series modeling as well^{50,51}. Building on these insights, we adopt a self-supervised framework with CI strategy in pre-training stage, where each variable is treated as an independent univariate sequence^{49,52}, as illustrated in Fig. 2a. This design prevents the model from prematurely relying on spurious correlations and facilitates the identification of distinctive dynamics within individual series. During the pre-training stage, only the multi-scale time-mixing module is trained in a self-supervised manner to learn noise-free temporal representations for each univariate sequence.

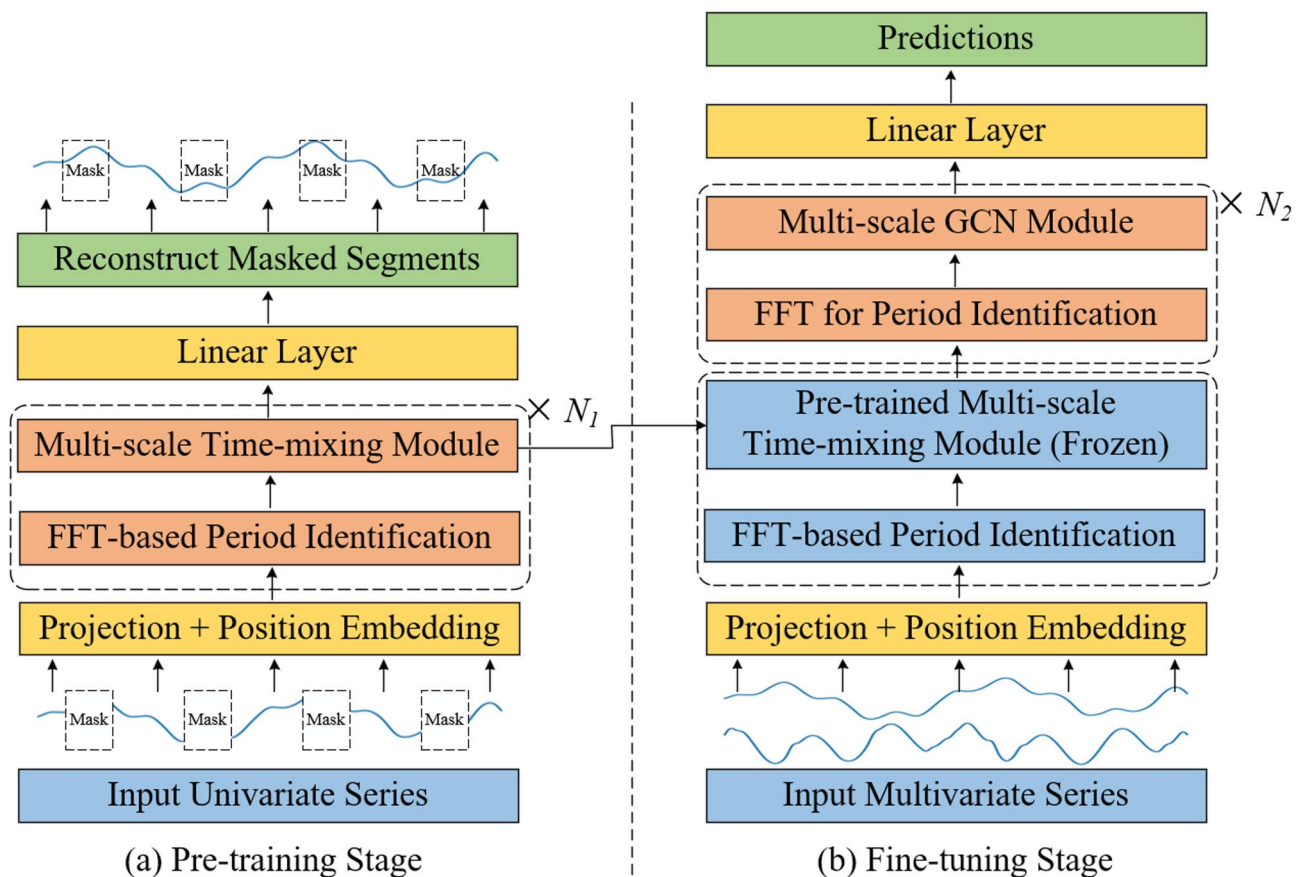


Fig. 2. The framework of the proposed PMSRWKV-GCN. **(a)** Channel-independent pre-training stage, where the multi-scale time-mixing module learns intra-series temporal dynamics through masked reconstruction. **(b)** Fine-tuning stage, where the pre-trained multi-scale time-mixing module (frozen) is integrated with a multi-scale GCN to capture intra-series temporal dependencies and inter-series spatial correlations across channels and scales.

In the subsequent fine-tuning stage, multivariate time series are fed into the model, where each channel is first processed by the pre-trained multi-scale time-mixing module to obtain clean temporal embeddings. These embeddings are then combined to reconstruct the multivariate representations, which are subsequently used to train the multi-scale GCN module, as illustrated in Fig. 2b. Through this two-stage design, the pre-trained temporal encoder provides denoised and informative temporal representation, while the GCN module captures the most informative intra-series temporal dependencies and inter-series spatial dependencies across multiple scales.

The self-supervised learning framework involves a masked reconstruction strategy, where portions of the input data are randomly masked and the model is trained to recover the missing segments. The overall pre-training process is summarized as follows:

Data masking

For each channel, multiple segments of fixed length are randomly masked within the input time window. Following the setup of PatchTST⁵⁰, the segment length and window size are set to 64 and 512, respectively. Masking is applied independently to each channel, consistent with the channel-independent design. The model is then required to reconstruct the masked values from their surrounding unmasked context, thereby encouraging the learning of intrinsic temporal structure within each univariate series.

Model pre-training

During the CI pre-training stage, only the multi-scale time-mixing module is trained to learn noise-free temporal representations for each individual channel. The module predicts the masked segments using only the unmasked portions of the same sequence, without relying on cross-channel information. The multi-scale GCN module is disabled at this stage since spatial dependencies are not yet introduced.

During fine-tuning, the pre-trained temporal encoder is kept frozen by design to preserve the channel-independent temporal representations learned from univariate (per-channel) pre-training. Only the multi-scale GCN is optimized to capture inter-series dependencies on multivariate inputs.

Loss calculation

The prediction accuracy is evaluated using the Mean Squared Error (MSE). The formula is given by:

$$MSE = \frac{1}{m} \sum_{j=1}^m \frac{1}{n_j} \sum_{i=1}^{n_j} (y_{ij} - \hat{y}_{ij})^2 \quad (28)$$

where y_{ij} and $\hat{y}_{ij}$ are the true and predicted values for the j -th masked segment, n_j is the length of the segment, and m denotes the total number of masked segments.

Computational complexity analysis

Computational efficiency is a central motivation for adopting RWKV, whose time-mixing mechanism avoids the quadratic cost of self-attention and achieves linear scalability with respect to sequence length. Since PMSRWKV-GCN extends RWKV with FFT-driven multi-scale time-mixing and multi-scale GCN, we analyze whether these extensions preserve the efficiency advantage of RWKV.

Let L denote the input sequence length, N the number of variables (channels), d the hidden dimension, k the number of dominant frequencies retained by FFT. We further denote N_1 and N_2 as the number of multi-scale time-mixing layers and the number of multi-scale GCN layers, respectively. We analyze the asymptotic time complexity of a forward pass, as the backward pass introduces only a constant-factor overhead that does not affect the overall complexity order.

In the self-supervised pre-training stage, PMSRWKV-GCN adopts a CI strategy and disables the GCN module, so the computation is dominated by temporal modeling within each channel. The multi-scale time-mixing module aggregates information from k period scales. Compared with the canonical RWKV time-mixing block, this introduces a multiplicative factor k while maintaining the linear dependence on L . Thus, for one channel, the time-mixing cost across N_1 layers is $O(N_1 \cdot k \cdot L \cdot d^2)$.

Since CI treats each channel independently, the overall temporal complexity scales linearly with the number of channels N , resulting in $O(N \cdot N_1 \cdot k \cdot L \cdot d^2)$. Additionally, FFT is applied independently to each channel to extract dominant frequencies, contributing $O(N \cdot L \log L)$. Therefore, the pre-training forward complexity is

$$T_{pre} = O(N \cdot N_1 \cdot k \cdot L \cdot d^2) + O(N \cdot L \log L) \quad (29)$$

This shows that the pre-training stage preserves linear scalability with respect to sequence length L .

In the fine-tuning stage, PMSRWKV-GCN introduces the multi-scale GCN module to capture inter-series dependencies. The temporal modeling component remains linear in L as above. During fine-tuning, FFT is computed once on the multivariate input window to obtain shared dominant periods across channels, contributing $O(L \log L)$.

The spatial modeling component operates on the variable dimension (nodes) rather than the time dimension. For a graph with E edges, one GCN layer typically costs $O(E \cdot d)$. Under a dense learned adjacency, $E = O(N^2)$. Considering k scales and N_2 stacked layers, the multi-scale GCN contributes $O(k \cdot N_2 \cdot E \cdot d)$.

In our design, the GCN operates on the channel representations (nodes) after temporal encoding, and is applied once per input window rather than per time step. Notably, this spatial term is independent of L . In addition, the dynamic adjacency matrix is updated using a self-attention mechanism across variables. Unlike Transformer attention which operates along the temporal dimension, this attention is computed across variables. Its complexity is $O(N^2 \cdot d)$, which is already included in the spatial modeling term above and remains independent of the sequence length L . Putting the components together, the fine-tuning (and thus inference) complexity is

$$T_{infer} = O(N \cdot N_1 \cdot k \cdot L \cdot d^2) + O(k \cdot N_2 \cdot E \cdot d) + O(L \log L) \quad (30)$$

This analysis shows that PMSRWKV-GCN preserves the linear temporal complexity of RWKV. Unlike attention-based models whose computational cost grows quadratically with the sequence length L , the additional multi-scale time-mixing and GCN components do not introduce any quadratic dependence on L . The temporal cost remains linear in L , while the extra computational overhead arises mainly from the variable dimension rather than the temporal dimension.

Experiments

This section evaluates the performance of the proposed PMSRWKV-GCN on multiple benchmark datasets under different forecasting horizons. The results are compared with representative baseline models to assess the accuracy. Performance is measured using MSE and Mean Absolute Error (MAE).

Datasets

We evaluated PMSRWKV-GCN on eight benchmark datasets, covering diverse temporal patterns and spatial correlations. Each dataset was chronologically divided into training, validation, and testing subsets. The split ratios follow standard practice 7 : 1 : 2 for the Weather, Traffic, and Electricity datasets, and 6 : 2 : 2 for the ETT series^{19,20}. The statistical summaries of all datasets are presented in Table 1.

- ETT (Electricity Transformer Temperature): Contains four subsets—two recorded hourly (ETTh1, ETTh2) and two at 15-minute intervals (ETTm1, ETTm2)—reflecting both long- and short-term periodicity.

Datasets	Electricity	Weather	ETTh1	ETTh2	ETTm1	ETTm2	Exchange	Traffic
Features	321	21	7	7	7	7	8	862
Timesteps	26,304	52,696	17,420	17,420	69,680	69,680	7588	17,544
Frequency	1 h	10 min	1 h	1 h	15 min	15 min	1 Day	1 h

Table 1. Statistical characteristics of datasets.

Hyper-parameter	Range
Number of layers (N_1)	[1, 2, 3]
Number of layers (N_2)	[1, 2, 3, 4]
d_{model}	[32, 64, 128, 256]
Node vector embedding	[24, 48, 96, 128]

Table 2. Search space of model hyperparameters.

Datasets	Weather	ETTh1	ETTh2	ETTm1	ETTm2	Electricity	Exchange	Traffic
Number of layers (N_1)	1	1	1	1	1	1	1	1
Number of layers (N_2)	2	2	2	2	2	3	2	3
d_{model}	32	32	32	32	32	256	64	256
Node vector embedding	24	24	24	24	24	96	24	96

Table 3. Dataset-specific hyperparameter configurations.

- Exchange: Comprises daily exchange rates among eight major economies, including New Zealand, the UK, China, Japan, Switzerland, Canada, Singapore, and Australia.
- Traffic: Reports highway occupancy rates collected from loop sensors in California, reflecting regional spatial dynamics.
- Electricity: Provides power consumption readings (kWh) from 321 clients, exhibiting strong diurnal and weekly regularities.
- Weather: Includes 21 meteorological variables such as air temperature, humidity, and wind speed. The measurements were taken every 10 min throughout 2020 in Germany.

Baselines and experimental setup

Baselines

To assess the performance of PMSRWKV-GCN, we compare it with six representative multivariate time series forecasting models spanning diverse architectural paradigms. The baselines include transformer-based models (Autoformer²⁰, Informer¹⁹, and ETSformer⁵³), graph-based models (MSGNet⁵⁴), linear approaches (DLinear⁵⁵), and period decomposition methods (TimesNet⁴⁷).

Pre-training stage

Following the protocol of mask-reconstruction self-supervised learning (Nie et al.⁴⁵), we perform CI pre-training for 100 epochs on each dataset. The look-back window and segment length are set to 512 and 64, respectively, and a masking ratio of 50% is applied with zero values used for the masked segments. Five dominant frequencies ($k=5$) are extracted using FFT following the frequency sparsity principle adopted in TimesNet⁴⁷. Real-world time series typically exhibit sparse spectra where only a few frequencies correspond to meaningful periodic structures, while higher-frequency components are often dominated by noise⁵⁶. Therefore, selecting the top- k dominant frequencies provides a stable structural prior for scale selection rather than a tunable hyperparameter.

To avoid any potential data leakage, the self-supervised pre-training is conducted strictly on the training split of each dataset. The validation and test sets are not involved in the pre-training process. The chronological order of the time series is strictly preserved, ensuring that no future information is introduced during pre-training.

Experimental setup

All experiments are implemented in PyTorch and executed on a workstation equipped with two NVIDIA GeForce RTX 2080 Ti GPUs. For fair comparison, the look-back window is fixed at $L=96$, corresponding to forecasting horizons of {96, 192, 336, 720}. The model is trained using the MSE loss with a batch size of 32 and an initial learning rate of 1×10^{-4} , and each experiment is repeated five times to ensure statistical reliability. Hyperparameters are tuned on the validation set via rolling validation. The hyperparameter search space and optimal configuration are summarized in Tables 2 and 3, respectively. For the CI pre-training stage, we found that using a single time-mixing layer achieves the best trade-off between stability and generalization across all datasets, while deeper configurations offer no additional benefits for univariate masked reconstruction. For

the multi-scale GCN module, deeper layers ($N_2=3$) are beneficial for datasets with large numbers of nodes (Electricity and Traffic), whereas smaller datasets perform better with shallower configurations ($N_2=2$). The results of comparison models are reported from TimesNet⁴⁷ and MSGNet⁵⁴.

Experimental comparison

Table 4 summarizes the comparative results of various benchmark models across eight datasets under multiple forecasting horizons {96, 192, 336, 720}. For each dataset and prediction length, the best result is boldfaced, and the second-best is underlined for clarity. These results consistently confirm the superior performance of the proposed PMSRWKV-GCN model. The entry “w/o Pretraining” denotes the same model trained directly without the pre-training phase.

From Table 4, several important observations can be made:

(1) The proposed PMSRWKV-GCN consistently outperforms competing approaches across diverse datasets and forecasting horizons. In most cases, it achieves the lowest average MSE and MAE, demonstrating a strong ability to model intricate temporal dependencies and cross-channel spatial correlations.

Models		Proposed		w/o Pretraining		MSGNet		TimesNet		DLinear		ETSformer		Autoformer		Informer	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Weather	96	0.155	0.187	<i>0.159</i>	<i>0.192</i>	0.163	0.212	0.172	0.220	0.196	0.255	0.197	0.281	0.266	0.336	0.300	0.384
	192	0.201	0.242	<i>0.207</i>	<i>0.250</i>	0.212	0.254	0.219	0.261	0.237	0.296	0.237	0.312	0.307	0.367	0.598	0.544
	336	0.253	0.267	<i>0.265</i>	<i>0.273</i>	0.272	0.299	0.280	0.306	0.283	0.335	0.298	0.353	0.359	0.395	0.578	0.523
	720	0.320	<i>0.318</i>	<i>0.322</i>	<i>0.330</i>	0.350	0.348	0.365	0.359	0.345	0.381	0.352	0.288	0.419	0.428	1.059	0.741
	Avg	0.232	0.254	<i>0.238</i>	<i>0.261</i>	0.249	0.278	0.259	0.287	0.265	0.317	0.271	0.309	0.338	0.382	0.634	0.548
ET'Tm1	96	0.293	0.330	<i>0.306</i>	<i>0.342</i>	0.319	0.366	0.338	0.375	0.345	0.372	0.375	0.398	0.505	0.475	0.672	0.571
	192	0.343	0.357	<i>0.351</i>	<i>0.371</i>	0.376	0.397	0.374	0.387	0.380	0.389	0.408	0.410	0.553	0.496	0.795	0.669
	336	0.377	0.398	<i>0.398</i>	<i>0.405</i>	0.417	0.422	0.410	0.411	0.413	0.413	0.435	0.428	0.621	0.537	1.212	0.871
	720	0.452	0.429	<i>0.461</i>	<i>0.432</i>	0.481	0.458	0.478	0.450	0.474	0.453	0.499	0.462	0.671	0.561	1.166	0.823
	Avg	0.366	0.379	<i>0.379</i>	<i>0.388</i>	0.398	0.411	0.400	0.406	0.403	0.407	0.429	0.425	0.588	0.517	0.961	0.734
ET'Tm2	96	0.162	0.247	<i>0.171</i>	<i>0.260</i>	0.177	0.262	0.187	0.267	0.193	0.292	0.189	0.280	0.255	0.339	0.365	0.453
	192	0.223	0.286	<i>0.235</i>	<i>0.293</i>	0.247	0.307	0.249	0.309	0.284	0.362	0.253	0.319	0.281	0.340	0.533	0.563
	336	0.275	0.306	<i>0.284</i>	<i>0.312</i>	0.312	0.346	0.321	0.351	0.369	0.427	0.314	0.357	0.339	0.372	1.363	0.887
	720	0.401	0.371	<i>0.407</i>	<i>0.384</i>	0.414	0.403	0.408	0.403	0.554	0.522	0.414	0.413	0.433	0.432	3.379	1.338
	Avg	0.265	0.303	<i>0.274</i>	<i>0.312</i>	0.288	0.330	0.291	0.333	0.350	0.401	0.293	0.342	0.327	0.371	1.410	0.810
ET'Th1	96	0.354	0.366	<i>0.367</i>	<i>0.388</i>	0.390	0.411	0.384	0.402	0.386	0.400	0.494	0.479	0.449	0.459	0.865	0.713
	192	0.419	0.392	<i>0.433</i>	<i>0.417</i>	0.442	0.442	0.436	0.429	0.437	0.432	0.538	0.504	0.500	0.482	1.008	0.792
	336	0.438	0.425	<i>0.453</i>	<i>0.441</i>	0.480	0.468	0.491	0.469	0.481	0.459	0.574	0.521	0.521	0.496	1.107	0.809
	720	0.460	0.463	<i>0.475</i>	<i>0.470</i>	0.494	0.488	0.521	0.500	0.519	0.516	0.562	0.535	0.514	0.512	1.181	0.865
	Avg	0.418	0.412	<i>0.432</i>	<i>0.429</i>	0.452	0.452	0.458	0.450	0.456	0.452	0.542	0.510	0.496	0.487	1.040	0.795
ET'Th2	96	0.282	0.34	<i>0.297</i>	<i>0.352</i>	0.328	0.371	0.340	0.374	0.333	0.387	0.340	0.391	0.346	0.388	3.755	1.525
	192	0.369	0.362	<i>0.388</i>	<i>0.385</i>	0.402	0.414	0.402	0.414	0.477	0.476	0.430	0.439	0.456	0.452	5.602	1.931
	336	0.416	0.411	<i>0.423</i>	<i>0.421</i>	0.435	0.443	0.452	0.452	0.594	0.541	0.485	0.479	0.482	0.486	4.721	1.835
	720	0.375	0.423	<i>0.391</i>	<i>0.430</i>	0.417	0.441	0.462	0.468	0.831	0.657	0.500	0.497	0.515	0.511	3.647	1.625
	Avg	0.361	0.384	<i>0.375</i>	<i>0.397</i>	0.396	0.417	0.414	0.427	0.559	0.515	0.439	0.452	0.450	0.459	4.431	1.729
Electricity	96	0.158	0.246	<i>0.160</i>	<i>0.254</i>	0.165	0.274	0.168	0.272	0.197	0.282	0.187	0.304	0.201	0.317	0.274	0.368
	192	0.161	0.265	<i>0.165</i>	<i>0.268</i>	0.184	0.292	0.184	0.289	0.196	0.285	0.199	0.315	0.222	0.334	0.296	0.386
	336	0.174	0.273	<i>0.180</i>	<i>0.281</i>	0.195	0.302	0.198	0.300	0.209	0.301	0.212	0.329	0.231	0.338	0.300	0.394
	720	0.192	0.29	<i>0.196</i>	<i>0.297</i>	0.231	0.332	0.220	0.320	0.245	0.333	0.233	0.345	0.254	0.361	0.373	0.439
	Avg	0.171	0.269	<i>0.175</i>	<i>0.275</i>	0.194	0.300	0.193	0.295	0.212	0.300	0.208	0.323	0.227	0.338	0.311	0.397
Traffic	96	0.543	0.288	<i>0.552</i>	<i>0.291</i>	0.591	0.320	0.593	0.321	0.650	0.396	0.607	0.392	0.613	0.388	0.719	0.391
	192	0.549	0.296	<i>0.566</i>	<i>0.304</i>	0.615	0.330	0.617	0.336	0.598	0.370	0.621	0.399	0.616	0.382	0.696	0.379
	336	0.572	0.298	<i>0.581</i>	<i>0.316</i>	0.622	0.333	0.629	0.336	0.605	0.373	0.622	0.396	0.622	0.337	0.777	0.420
	720	0.582	0.313	<i>0.593</i>	<i>0.320</i>	0.634	0.346	0.640	0.350	0.645	0.394	0.632	0.396	0.660	0.408	0.864	0.472
	Avg	0.562	0.299	<i>0.573</i>	<i>0.308</i>	0.616	0.332	0.620	0.336	0.625	0.383	0.621	0.396	0.628	0.379	0.764	0.416
Exchange	96	0.097	<i>0.206</i>	0.103	0.222	0.102	0.230	0.107	0.234	<i>0.088</i>	0.218	0.085	0.204	0.197	0.323	0.847	0.752
	192	0.172	0.295	0.178	<i>0.302</i>	0.195	0.317	0.226	0.344	<i>0.176</i>	0.315	0.182	0.303	0.300	0.369	1.204	0.895
	336	<i>0.340</i>	0.403	0.347	<i>0.410</i>	0.359	0.436	0.367	0.448	0.313	0.427	0.348	0.428	0.509	0.524	1.672	1.036
	720	<i>0.851</i>	0.689	0.886	0.711	0.940	0.738	0.964	0.746	0.839	0.695	1.025	0.774	1.447	0.941	2.478	1.310
	Avg	<i>0.365</i>	0.398	0.379	<i>0.411</i>	0.399	0.430	0.416	0.443	0.354	0.414	0.410	0.427	0.613	0.539	1.550	0.998

Table 4. Forecasting results on eight datasets for comparison models.

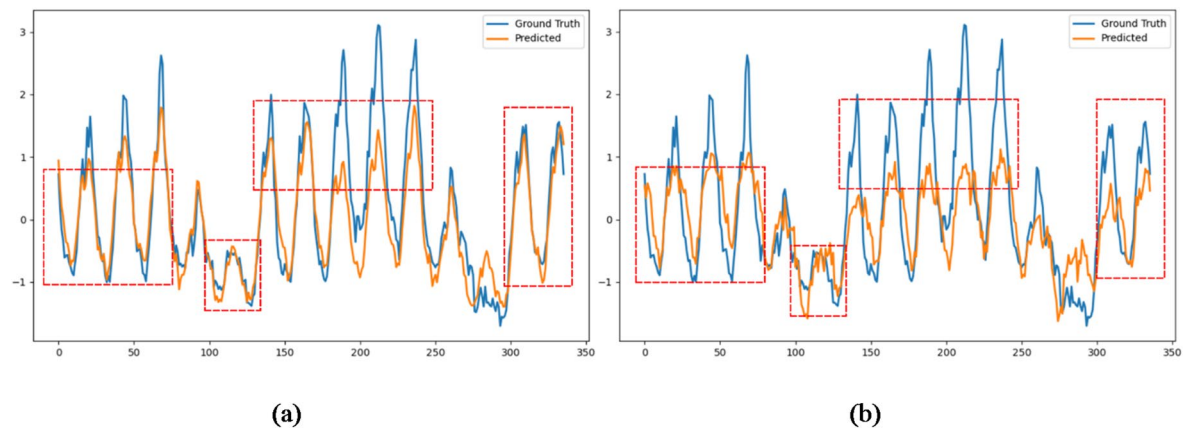


Fig. 3. Comparison between PMSRWKV-GCN and MSGNet on the Electricity dataset (prediction horizon = 336): (a) PMSRWKV-GCN; (b) MSGNet.

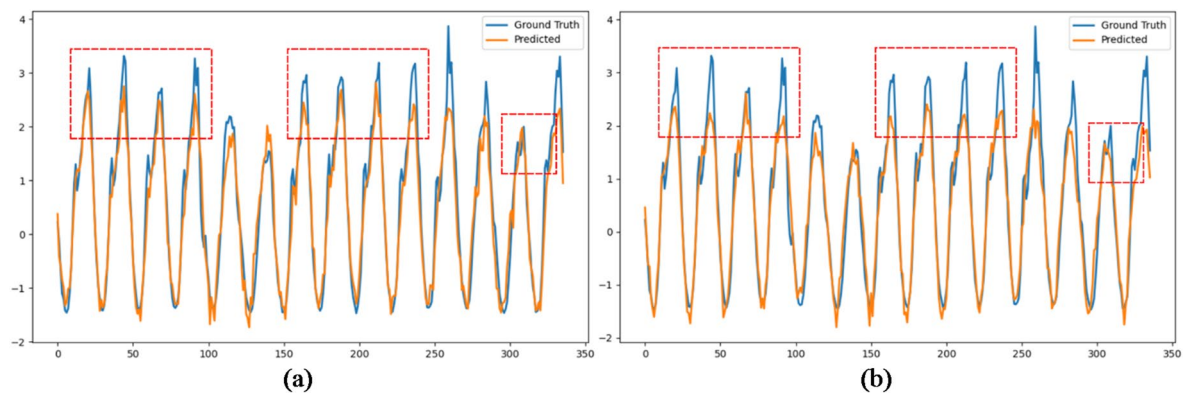


Fig. 4. Comparison between PMSRWKV-GCN and MSGNet on the Traffic dataset (prediction horizon = 336): (a) PMSRWKV-GCN; (b) MSGNet.

(2) Even without the pre-training phase (denoted as w/o Pretraining), the model remains highly competitive, indicating strong intrinsic generalization ability. Nevertheless, pre-training provides additional benefits, particularly in stabilizing optimization and improving overall accuracy.

(3) On the Weather dataset, PMSRWKV-GCN surpasses MSGNet, reducing the average MSE and MAE by 6.83% ($0.249 \rightarrow 0.232$) and 8.63% ($0.278 \rightarrow 0.254$), respectively.

(4) On the Electricity dataset, it exhibits even stronger gains, reducing the two metrics by 11.86% ($0.194 \rightarrow 0.171$) and 10.33% ($0.300 \rightarrow 0.269$), respectively.

(5) For Exchange dataset, although the advantage diminishes at longer horizons, the average performance (MSE=0.365, MAE=0.398) remains competitive, revealing good adaptability to heterogeneous temporal patterns.

Overall, the proposed PMSRWKV-GCN demonstrates consistent advantages across datasets, underscoring the synergy between multi-scale temporal modeling and graph-based spatial learning.

Visual analysis of model predictions

Figures 3 and 4 present the prediction results of PMSRWKV-GCN and MSGNet on the Electricity and Traffic datasets, respectively. These figures depict the predicted values against the ground truth for a prediction length of 336. The predicted curves of PMSRWKV-GCN align more closely with the ground truth, particularly during rapid fluctuations (highlighted in the red-boxed regions), demonstrating its stronger capability in tracking temporal variations in highly dynamic time series.

Figure 5 further validates these findings by comparing the error distributions of PMSRWKV-GCN and MSGNet using violin plots. For both the Electricity and Traffic datasets, PMSRWKV-GCN exhibits errors that are more tightly concentrated around zero, indicating more stable and accurate predictions compared with MSGNet. The narrower error distribution also suggests that PMSRWKV-GCN experiences fewer extreme deviations, which is essential for reliable and stable forecasting.

In summary, the visual analyses further confirm that PMSRWKV-GCN delivers more stable and accurate forecasts across diverse datasets, particularly in scenarios with pronounced temporal variability.

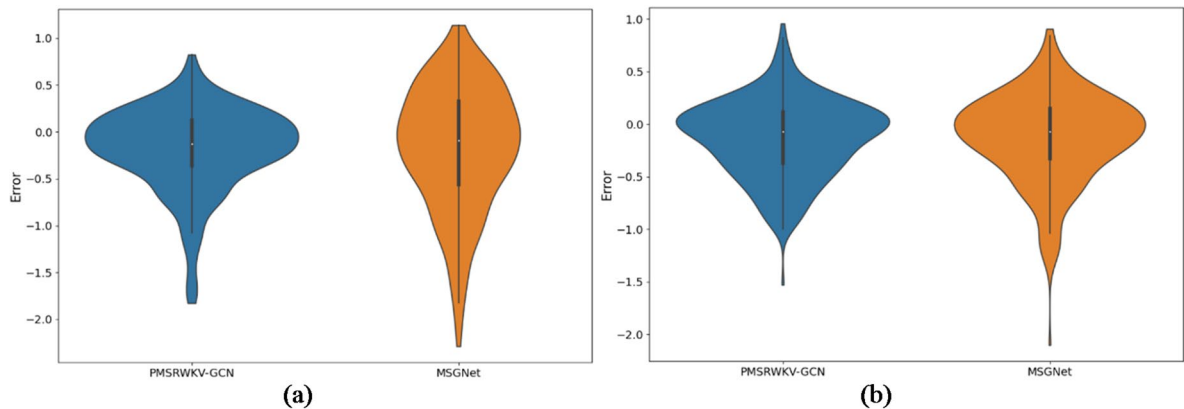


Fig. 5. Violin plots of prediction errors on the Electricity and Traffic datasets (prediction horizon = 336): (a) Electricity; (b) Traffic.

Datasets	Weather		ETTh1		ETTh2		ETTh1		ETTh2		Electricity		Traffic		Exchange	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
PMSRWKV-GCN	0.232	0.254	0.366	0.379	0.265	0.303	0.418	0.412	0.361	0.384	0.171	0.269	0.562	0.299	0.365	0.398
w/o Pretraining	0.238	0.261	0.379	0.388	0.274	0.312	0.432	0.429	0.375	0.397	0.175	0.275	0.573	0.308	0.379	0.411
Improvement	2.52%	2.68%	3.43%	2.32%	3.28%	2.88%	3.24%	3.96%	3.73%	3.27%	2.29%	2.18%	1.92%	2.92%	3.69%	3.16%

Table 5. Impact of pre-training on model performance.

Datasets		Weather	ETTh1	Electricity	Traffic	Exchange
w/o Pre-training	MSE	0.322	0.461	0.196	0.593	0.886
	MAE	0.330	0.432	0.297	0.320	0.711
wo/ Multi-scale Time-mixing	MSE	0.355	0.480	0.241	0.644	0.932
	MAE	0.350	0.462	0.353	0.347	0.735
wo/ Multi-scale GCN	MSE	0.407	0.622	0.324	0.853	1.261
	MAE	0.411	0.539	0.420	0.433	0.967

Table 6. Ablation study on modules of the proposed model without pre-training (prediction horizon = 720).

Ablation studies

Effect of pre-training

Pre-training serves as an essential component for enhancing the performance of the proposed PMSRWKV-GCN. Table 5 reports the ablation results evaluating the impact of the pre-training phase. The averaged MSE and MAE over four prediction lengths {96, 192, 336, 720} are provided for each dataset, along with the corresponding relative improvements after pre-training.

The results show that the pre-training consistency enhances model performance across all datasets. The average MSE reduction ranges from 1.92% to 3.73%, indicating that pre-training enables the model to capture temporal structures more effectively before fine-tuning. This improvement remains consistent across datasets of different domains, including electricity consumption (ETT, Electricity), traffic flow (Traffic), and currency exchange rates (Exchange). These findings suggest that pre-training helps the model converge to a better local optimum by providing cleaner and more informative temporal representations, thereby yielding more accurate and stable forecasts.

Module-wise ablation study (without pre-training)

To further examine the contribution of each component in PMSRWKV-GCN, we conduct a module-wise ablation study without the pre-training phase. This analysis isolates the individual effects of the multi-scale time-mixing module and the multi-scale GCN module. Table 6 summarizes the performance degradation when each module is removed.

The results indicates that both components are indispensable to the overall effectiveness of the model, as removing either module leads to a notable increase in error metrics:

- **wo/ multi-scale time-mixing:** Removing the multi-scale time-mixing module causes a clear decline in performance, as the full model achieves 3.96%–18.67% lower MSE compared to the version without this module.

This highlights the time-mixing module's essential role in modeling temporal dependencies and producing accurate predictions.

- **w/o/ multi-scale GCN:** The multi-scale GCN contributes even more substantially, particularly on datasets with complex spatial dependencies such as Traffic and Exchange. The full model achieves 15.65% to 39.51% lower MSE compared to the version without this module, underscoring the importance of this module in effectively capturing spatial correlations across multiple time scales.

Overall, the module-wise ablation study confirms that both modules are essential, with the multi-scale GCN playing a particularly important role in modeling inter-channel spatial interactions.

Recent studies such as PatchTST⁵⁰ and iTransformer²³ have shown that very competitive performance can be achieved in multivariate time series forecasting without relying on explicit graph-based spatial modeling, either through channel-independent temporal modeling (PatchTST⁵⁰ or through effective channel-mixing mechanisms (iTransformer²³). This motivates examining whether spatial modeling can still provide additional benefits within our framework.

As shown in Table 6, introducing the multi-scale GCN consistently improves PMSRWKV across all datasets, with MSE reductions ranging from 15.65% to 39.51%. This provides direct evidence that explicitly modeling inter-series dependencies via graph structure brings additional benefits beyond temporal modeling alone, particularly for datasets with complex spatial correlations such as Traffic and Exchange.

Conclusions

In this study, we introduced PMSRWKV-GCN, a novel model for multivariate time series forecasting that integrates a multi-scale time-mixing module with a multi-scale GCN. This unified design enables the model to capture intra-series temporal dependencies and inter-series spatial correlations in a multi-scale manner. By incorporating periodic cues through FFT and spatial interactions via GCN, the model effectively represents complex temporal patterns and cross-channel dynamics. Extensive experiments demonstrate that the proposed CI pre-training stage consistently improves forecasting accuracy by providing stable and noise-free temporal representations before fine-tuning. Ablation studies further reveal that both the multi-scale time-mixing and multi-scale GCN modules are essential components, as removing either leads to clear performance degradation—particularly on datasets with strong inter-series dependencies. Notably, even without pre-training, PMSRWKV-GCN maintains competitive performance, reflecting strong model stability and generalization ability.

Overall, PMSRWKV-GCN delivers a significant advancement in multivariate forecasting by jointly leveraging multi-scale temporal modeling and graph-based spatial learning. Building upon these findings, future research may explore incorporating external covariates (e.g., weather variables, economic indicators), enhancing interpretability to better understand factor-wise influences, and developing more adaptive adjacency learning mechanisms. Additionally, extending the framework to domain-specific applications such as energy forecasting or traffic management presents a promising direction toward more transparent, robust, and trustworthy time series forecasting.

Data availability

All datasets used in this study are publicly available.

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Jianhua Hao: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. Fangai Liu: Supervision, Project administration, Writing – review & editing. Weiwei Zhang: Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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